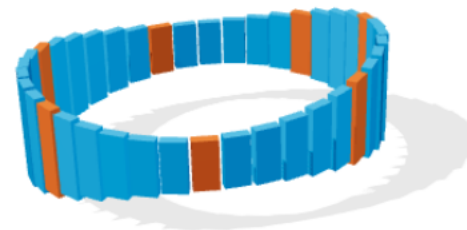


VLA- JEPA

Can watching
become doing?

Enhancing vision-language-action models with latent world prediction

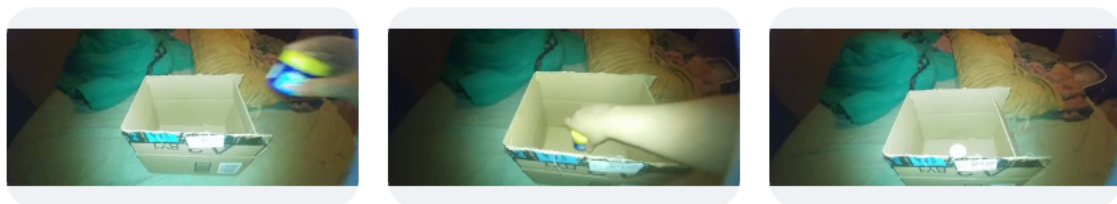


Transition → state → control

Video shows outcomes. Robot data supplies control.

WATCH

Human video



Before

During

After



“Put the jar into the box.”

Video + description; no robot control labels

ACT

Robot demonstration



“Place the green block inside the black bowl.”

Recorded pose commands · x/y/z normalized per axis



Same ambition. Different supervision.

Robot-labelled VLA

Recorded controls

Image + instruction → executable actions

LAPA

Future pixels

Learn video codes, then adapt them to control

UniVLA

DINO features

Feature reconstruction already existed

VLA-JEPA

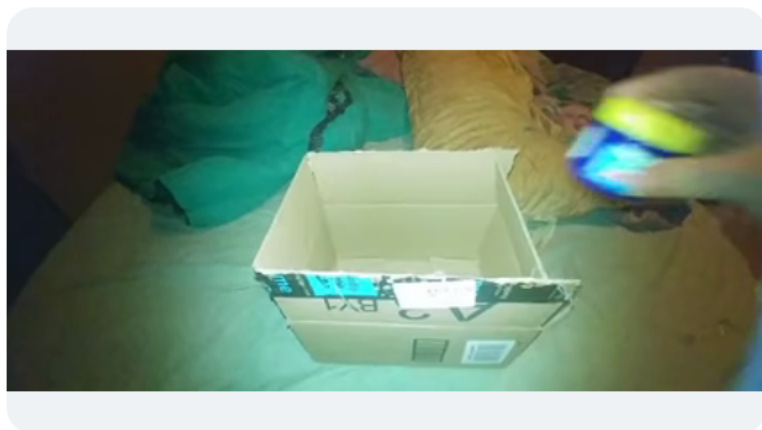
Future state features

Train policy tokens through predictive alignment

The key question: what does the policy see, and predict?

Predict a future state — in feature space

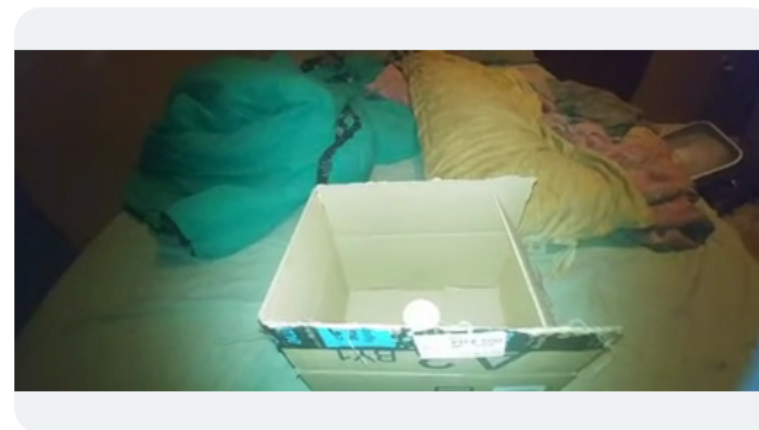
Now



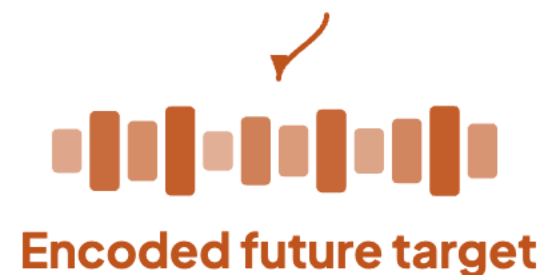
“Put the jar into the box”

Observed video transition

Later



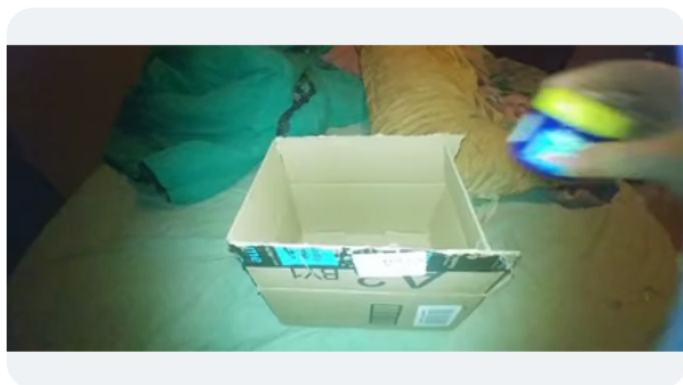
Compare



Like anticipating “jar inside box” — without drawing the next frame.

The tokens describe an intended transition

Current observation



Put the jar into the box

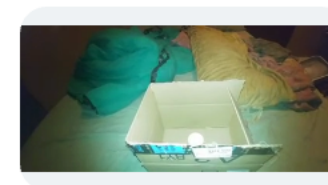
VLM

Qwen3-VL-2B



Latent action tokens

An intended transition

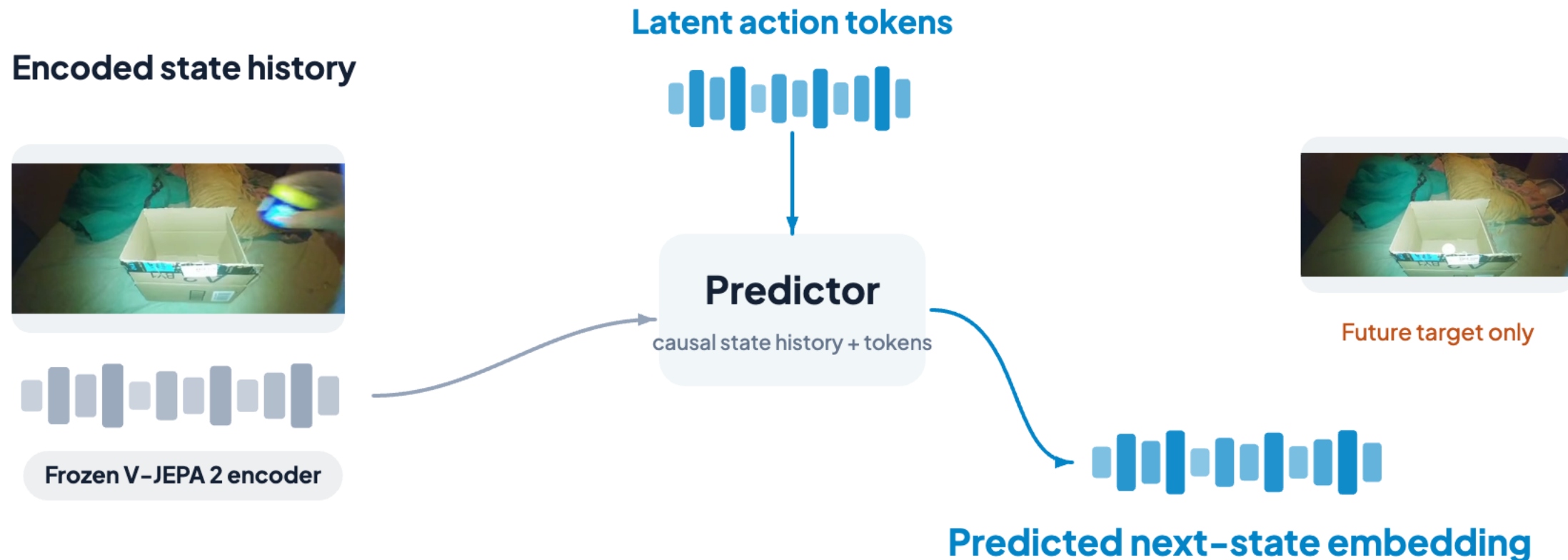


Supervision only

These tokens condition both prediction and control.

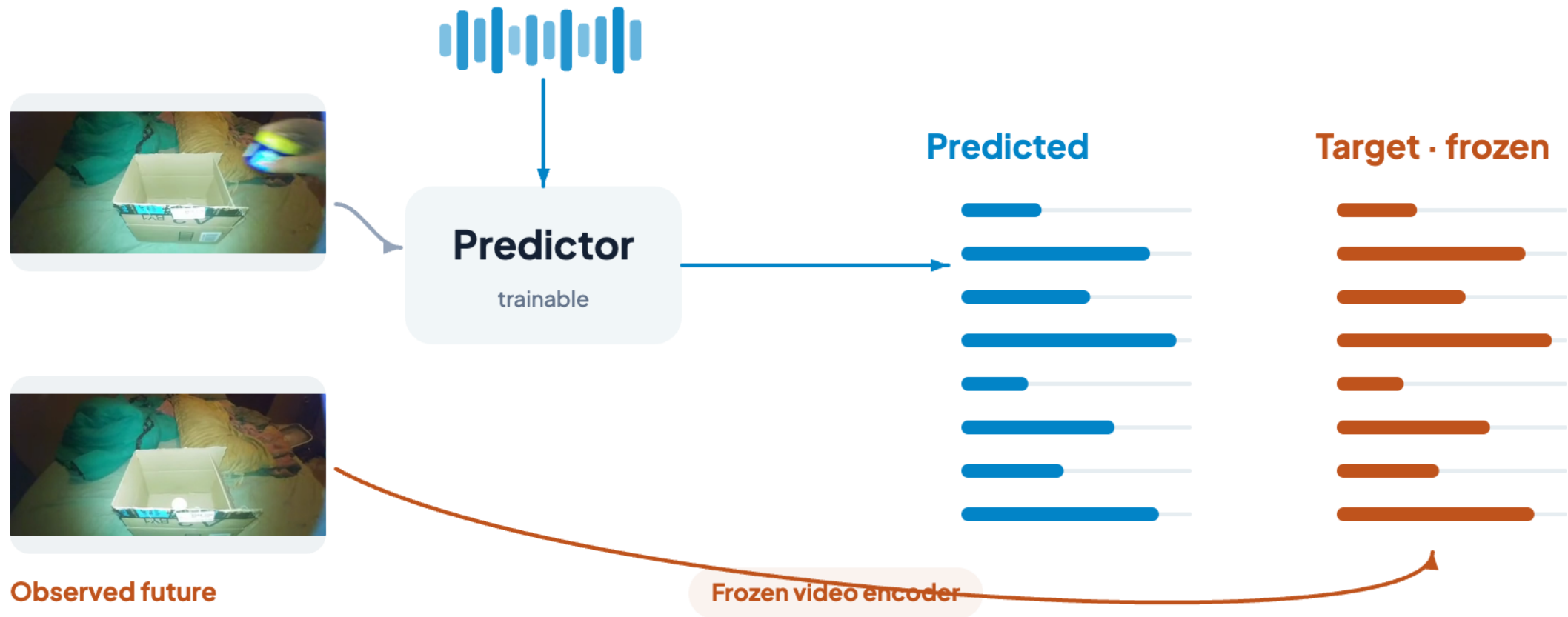
They are neither pixels nor executable motor commands.

The predictor outputs the next state's features



The output describes the next state; the tokens describe the transition.
No image decoder is needed for this supervision objective.

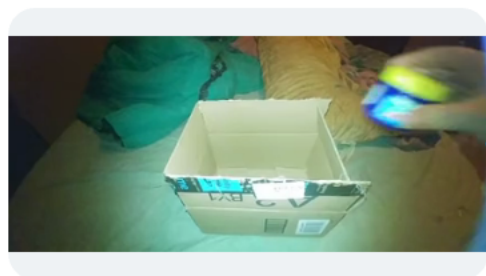
The target stays fixed. The prediction learns.



Training makes noisy corrections, not a literal motion between states.

The policy never receives its future target

Initial observation



Policy → latent tokens



Predictor

causal history

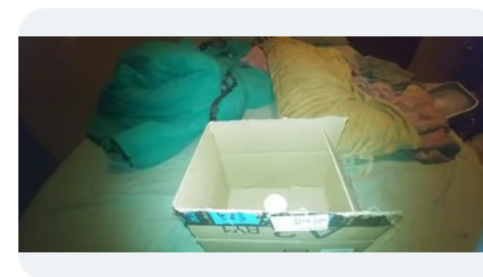
Prediction



Training target only

Future frames stay below this boundary.

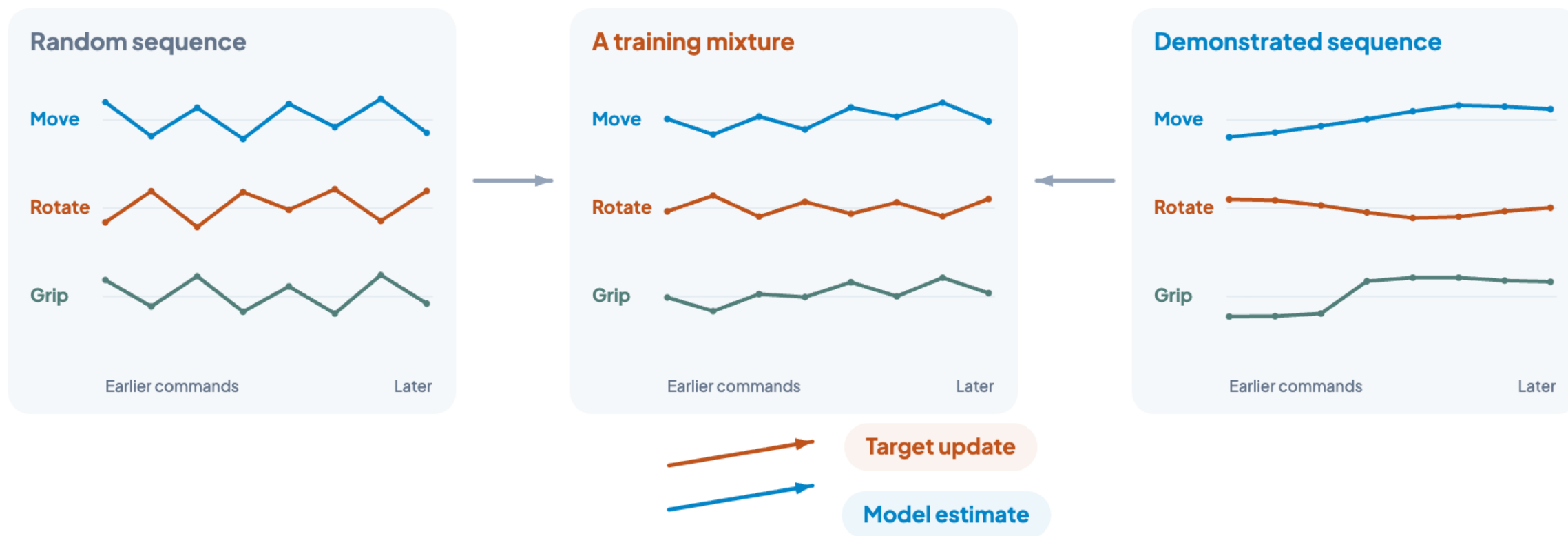
Compare only



At deployment: observation + instruction → tokens → action head.

Learn how to update a sequence of controls

Predict the update; match its training target



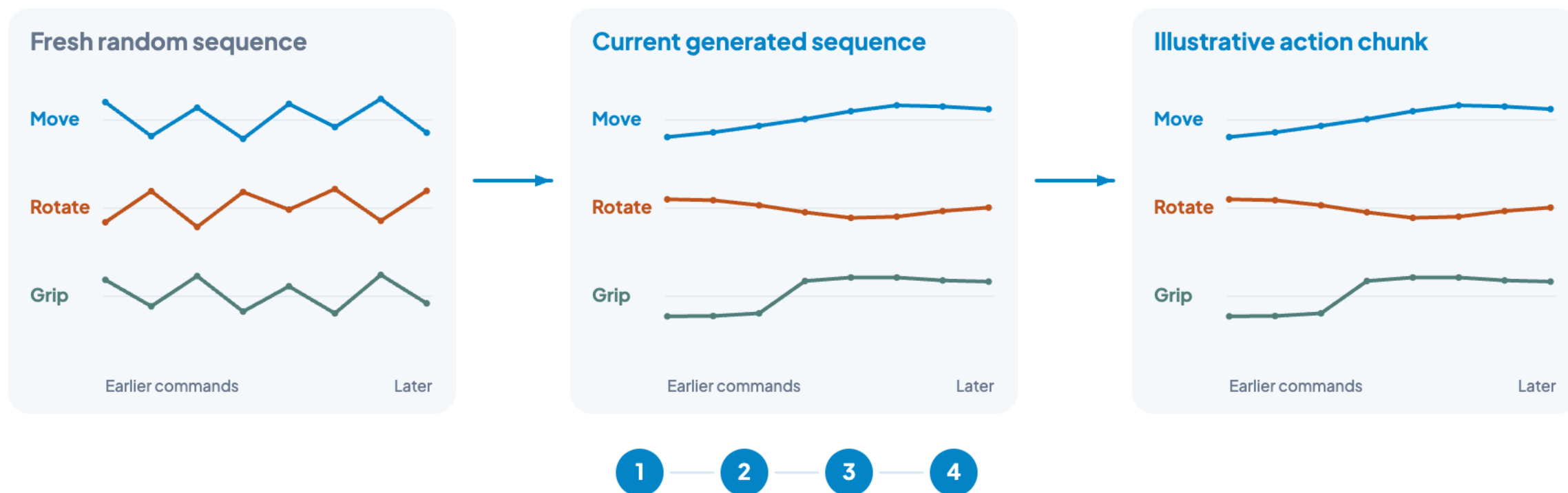
Conditioning: observation + instruction → policy tokens

These curves describe controls over time, not a robot's spatial trajectory.

Paper Eqs. 7–8 · Illustrative controls · Mixing time is not physical robot time

Turn a random sequence into an action chunk

Execute the generated controls



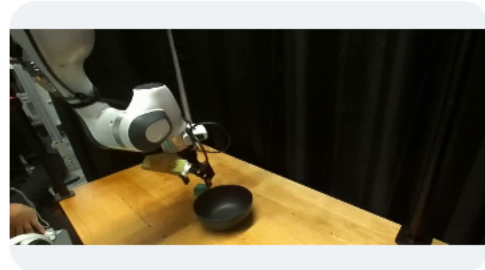
The model sees the task and generation time at every update.

Generation time is separate from the time axis inside the control chunk.

Video teaches prediction. Robot data teaches control.



Human video

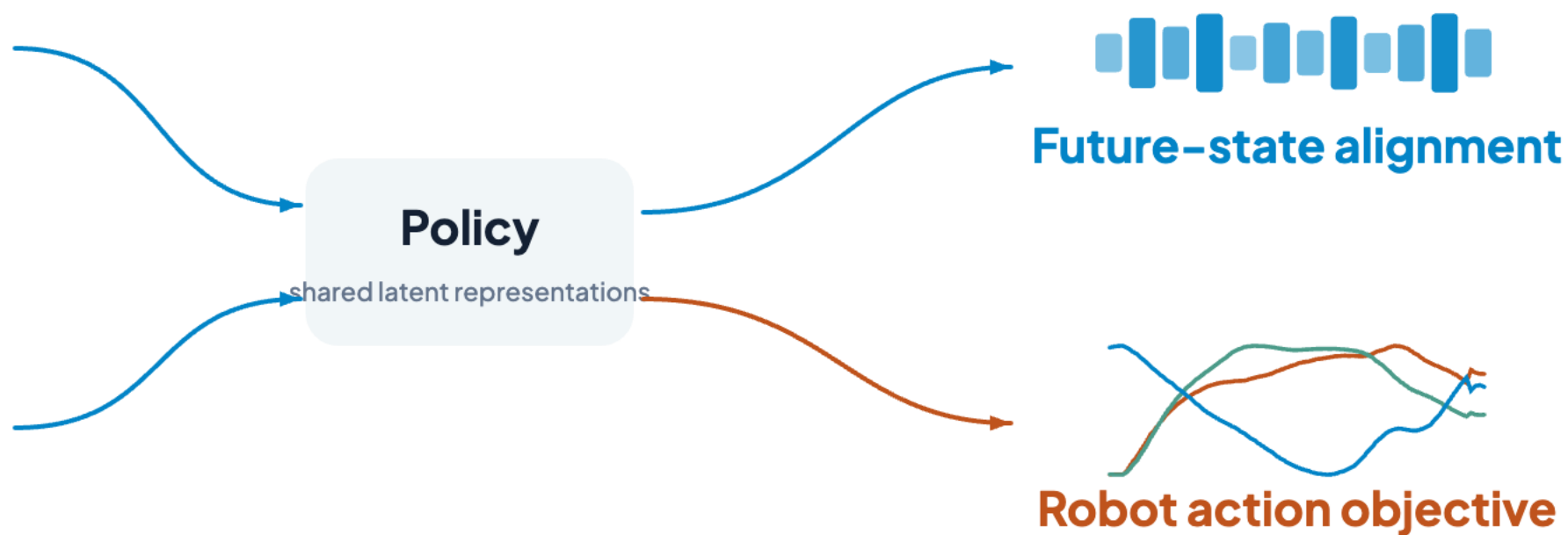


Robot demonstration

Different robots need compatible controls and target post-training.

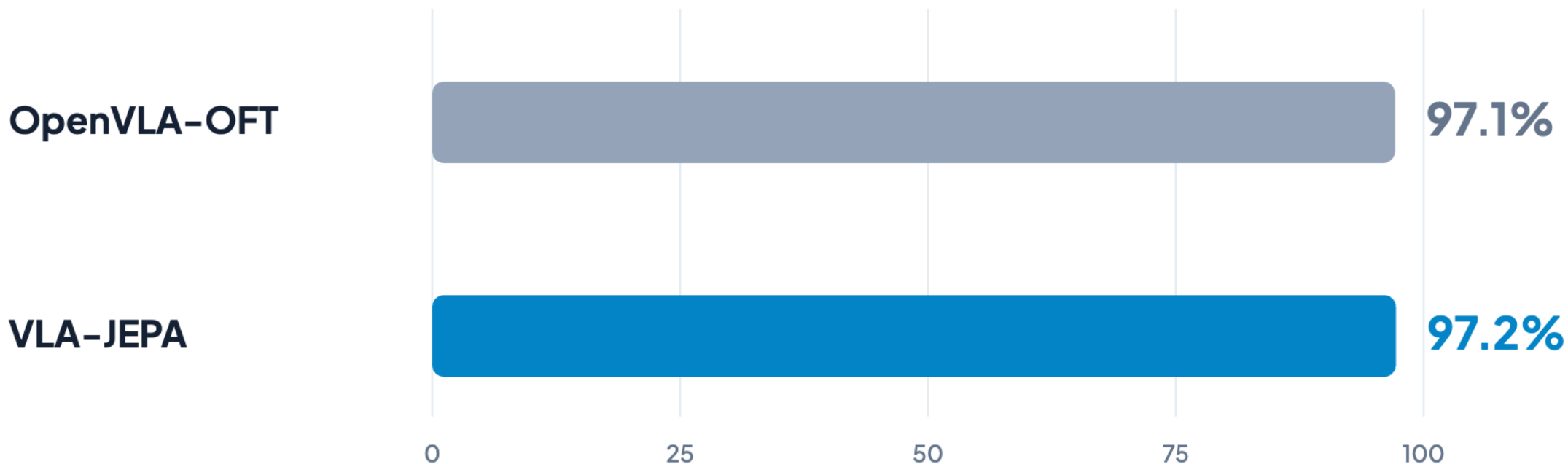
Coordinate frames · units · gripper conventions · command timing

Paper §3.3 · Human videos have no robot action labels · Robot demonstrations provide both objectives



Standard LIBERO: a near tie

Success rate



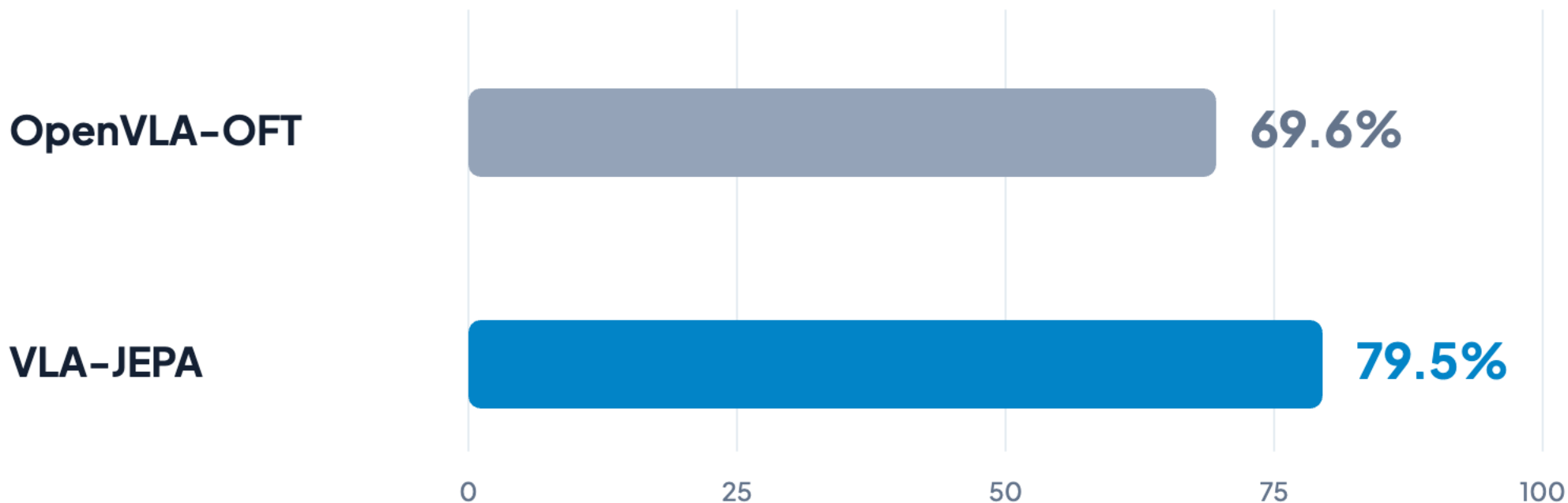
+0.1 percentage points

A near tie on standard tasks does not establish a meaningful advantage.

Paper Table 1 · Success rate (%) · No reported confidence intervals

LIBERO-Plus: stronger under perturbations

Success rate

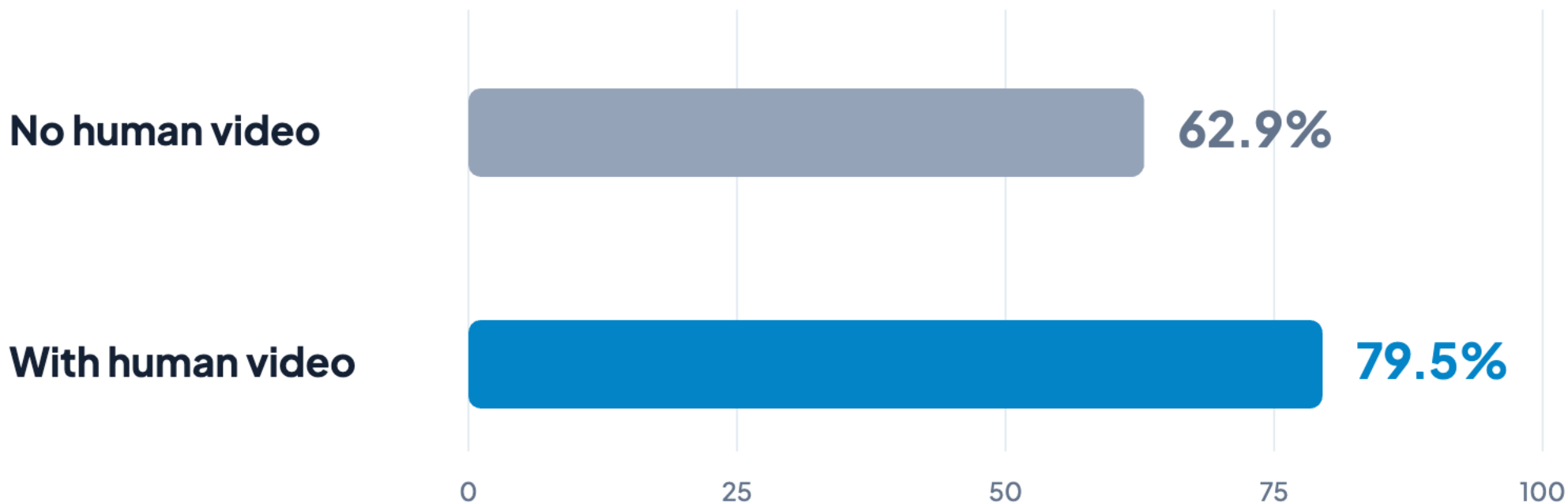


+9.9 percentage points

The stronger claim is resilience under changed conditions.

Human video improves LIBERO-Plus robustness

Success rate

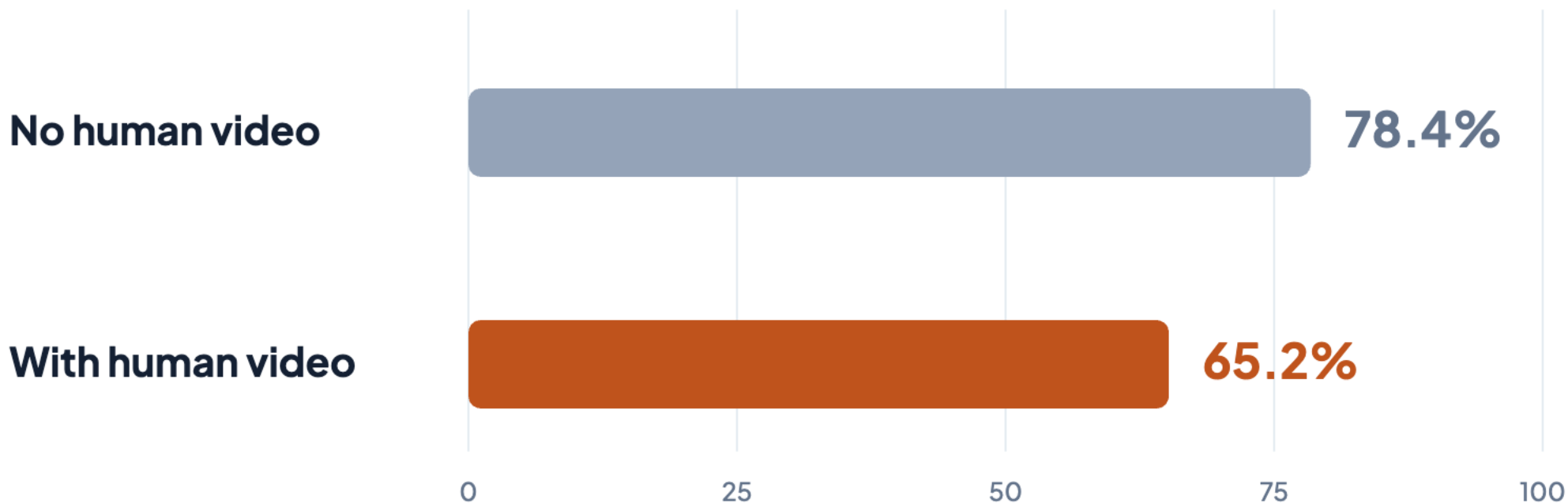


+16.6 percentage points

Compare the same method, with and without the extra video data.

Human video does not improve every task

Success rate



-13.2 percentage points · SimplerEnv Google

The benefit depends on the benchmark; human video is not uniformly helpful.

Can we lower the cost and keep useful control?

01

Reduced precision

Compare against the same baseline

Measure memory, latency and success

02

Smaller backbone

Explore SmoIVLM integration

Keep prediction and action components

03

Work so far

Implementation and evaluation setup

Validate before drawing conclusions

Matched baseline → resource use + task success

The trade-off is a question to test, not an improvement to promise.

Strong averages leave deployment questions open

SENSOR NOISE

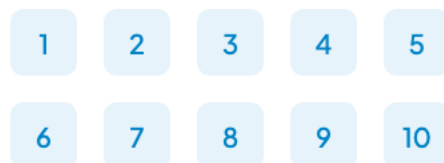
66.3%

VLA-JEPA

79.0% · pi zero

A robustness counterexample

REAL-WORLD EVIDENCE



10 trials per task

Wrong-object selections

TRAINING → DEPLOYMENT



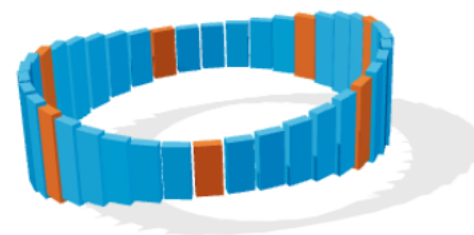
8 A100 GPUs · paper training

Proposed study: single-GPU efficiency

Headline averages do not establish deployment reliability.

No reported confidence intervals on these headline comparisons.

Predict states. Learn control.



Strongest evidence

Robustness under perturbations

Our open question: lower resource cost
while keeping useful control?

A smaller backbone is a hypothesis to evaluate



Keep the video encoder, world predictor and action head.
Check integration and matched evaluation before concluding.

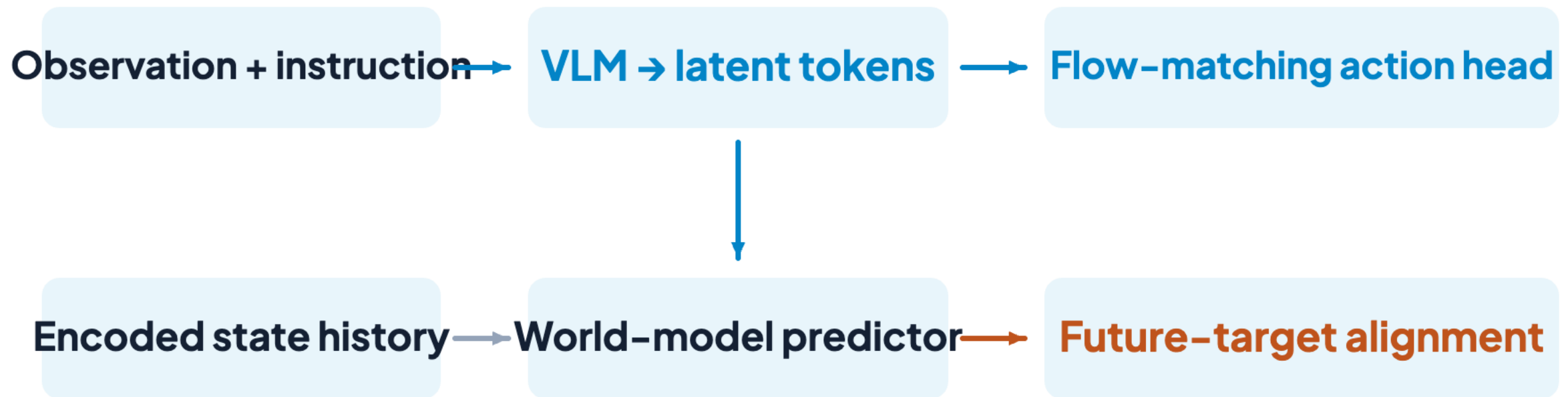
Architecture search comes after a reliable comparison



An optional direction, with an open outcome

Resource savings would need evidence of useful robot control.

The whole method, after the components are familiar



Video supervises prediction. Robot demonstrations also supervise actions.

The equations behind the visual training pair

Mix noise and a demonstrated action chunk

$$a_{\tau} = (1 - \tau)\epsilon + \tau a^*$$

Regress velocity onto the displacement of the pair

$$\mathcal{L}_{\text{FM}} = \mathbb{E} \left[\left\| v_{\theta}(a_{\tau}, \tau | z_a) - (a^* - \epsilon) \right\|_2^2 \right]$$

Mixing time describes generation progress, not physical robot time.